

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : D15

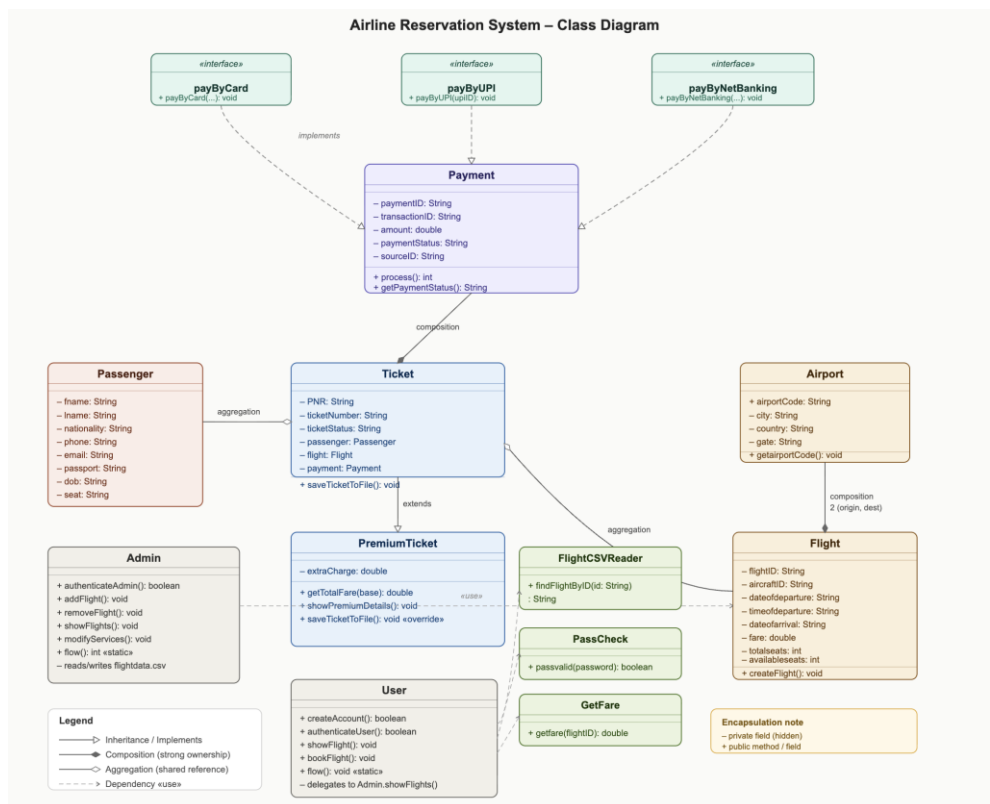
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1. Problem Title: **Airline Management System**

STRUCTURAL DESIGN

2. Class diagram



BEHAVIOURAL DESIGN

3. Use-case diagram

1. List of actors

1. **User / Passenger**

A customer who creates an account, logs in, views flights, books tickets, and makes payments.

2. **Admin**

The administrator who manages flights by adding, removing, viewing, and modifying airline services.

3. **Payment System**

External payment handling system used to process Card, UPI, and Net Banking transactions.

2. List of use-cases

Passenger Use Cases:

1. **Create Account**

Allows a new user to register by entering username, password, phone number, and email details.

2. **Login User**

Enables registered users to securely log into the system using username and password.

3. **View Flights**

Displays all available flights stored in the flight database.

4. **Book Flight**

Allows users to select a flight, enter passenger details, and reserve a seat.

5. **Process Payment**

Handles payment through Card, UPI, or Net Banking before confirming booking.

6. **Update Available Seats**

Automatically decreases the available seat count after successful booking.

7. **Admin Authentication**

Verifies administrator credentials before granting access to admin operations.

8. **Add Flight**

Allows the admin to add new flight details into the system database.

9. **Remove Flight**

Allows the admin to remove existing flights from the database.

10. **Modify Services**

Provides functionality for modifying airline services and facilities.

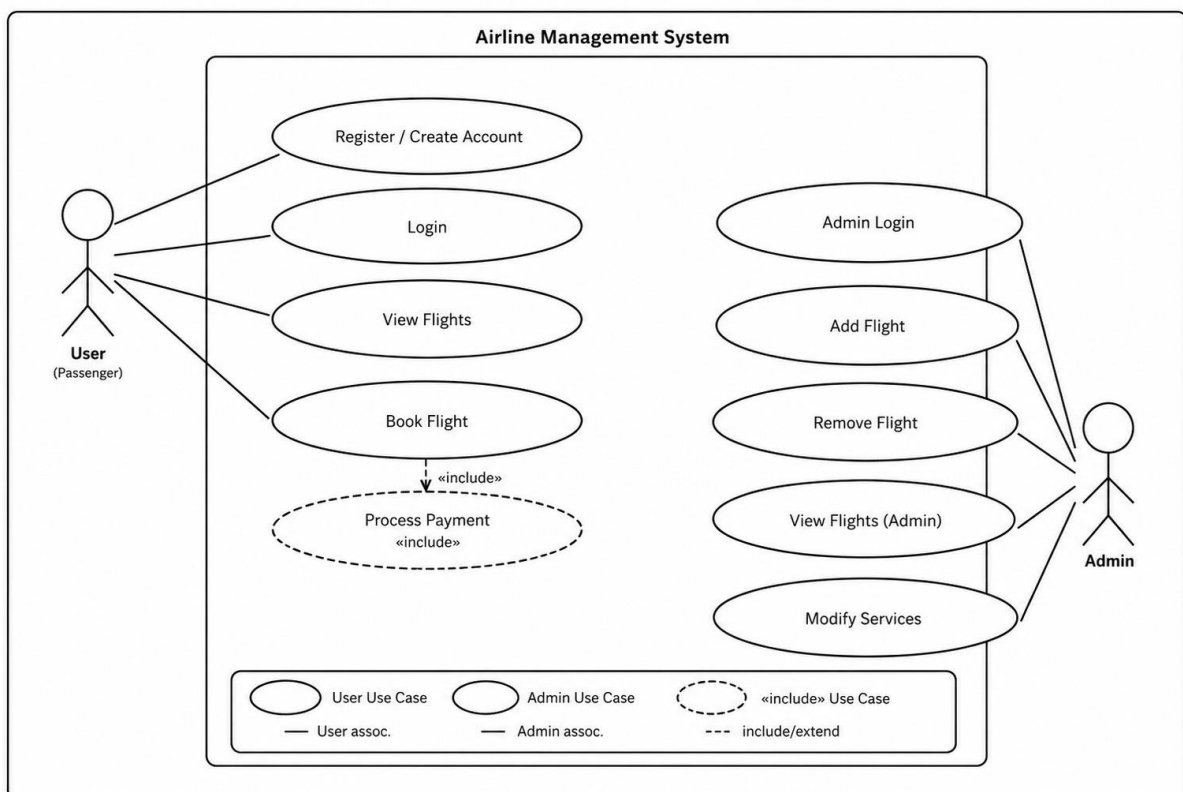
11. **View Flight Details**

Allows the admin to display all stored flight information.

Admin Use Cases

1. Add Flight: Allows administrators to enter new flight details into the system.
2. Update Flight Schedule: Enable the management of departure /arrival times and route information.
3. Cancel Flight: Allows the admin to remove a flight from the schedule due operational reasons.
4. Generate Reports: Produces operational and activity logs for the airline for the airline.
5. Customer Service: Monitor the status of passenger issues and queries and follow up on the resolution progress within the system.
6. Cabin Service: Oversee the fulfilment of meal requests initiated by passengers during the flight.

3. Use case diagram



4. Sequence diagram (use UML notations)

The flight booking process is one of the most important functionalities of the Airline Reservation System. Initially, the passenger logs into the system using valid credentials. After successful authentication, the user requests to view the available flights. The system retrieves flight details from the flight database and displays them to the user.

The passenger then selects a desired flight by entering the Flight ID. The system validates the entered Flight ID using the FlightCSVReader utility class and checks seat availability. If seats are available, the system asks the passenger to enter personal details such as first name, last name, nationality, phone number, email address, passport number, and date of birth.

After collecting passenger information, the system calculates the ticket fare and redirects the user to the payment module. The user chooses a payment method such as Card, UPI, or Net Banking, and the payment is processed successfully. Once the payment is completed, the system updates the available seat count in the flight database and confirms the booking. Finally, the passenger receives a booking confirmation message.

